

Part A Multiple choice (15 points)

1-5 ~~DBCE~~EC

6-10 DDCAD

11-15 BAE

Part B Short questions (15 points)

14. (a) cis isomer

(b) cis isomer, trans isomer

15.  $[-114^{\circ}\text{C} - (-155^{\circ}\text{C})] \times 42.0\text{g} \times 0.970 \text{ J/g}\cdot\text{K}$

$$+ \frac{42.0\text{g}}{46.0\text{g/mol}} \times 5.02 \times 10^3 \text{ J/mol}$$

$$+ [78^{\circ} - (-114^{\circ}\text{C})] \times 42.0\text{g} \times 2.30 \text{ J/g}\cdot\text{K}$$

$$+ \frac{42.0\text{g}}{46.0\text{g/mol}} \times 38.56 \times 10^3 \text{ J/mol}$$

$$= 6.00 \times 10^4 \text{ J}$$

16.  $\text{CF}_4$ ,  $\text{CClF}_3$ ,  $\text{CCl}_2\text{F}_2$ ,  $\text{CCl}_3\text{F}$

17. (a) gas  $\rightarrow$  solid  $\rightarrow$  liquid

(b) gas  $\rightarrow$  liquid  $\rightarrow$  solid

18. diethyl ether, acetone, ethanol, water.